# MSG No. 138.528 $P_c4_2/ncm$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `-4'm'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[1,6,7,8,26,27,28,29]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[2,3,4,5,25,30,31,32]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[9,14,15,16,18,19,20,21]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[10,11,12,13,17,22,23,24]` |

Table 2: Wyckoff site: `4b`, site symmetry: `-4m'2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[1,6,10,11,23,24,28,29]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,3,9,14,20,21,31,32]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[4,5,15,16,18,19,25,30]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[7,8,12,13,17,22,26,27]` |

Table 3: Wyckoff site: `4c`, site symmetry: `4'm'm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[1,6,15,16,18,19,28,29]` |
| 2 | $\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | `[2,3,12,13,17,22,31,32]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$ | `[4,5,10,11,23,24,25,30]` |
| 4 | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | `[7,8,9,14,20,21,26,27]` |

Table 4: Wyckoff site: `8d`, site symmetry: `..2/m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,8,9,16]` |
| 2 | $\left[\frac{1}{2}, 0, 0\right]$ | `[2,4,10,12]` |
| 3 | $\left[0, \frac{1}{2}, 0\right]$ | `[3,5,11,13]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[6,7,14,15]` |
| 5 | $[0, 0, 0]$ | `[17,24,25,32]` |
| 6 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[18,20,26,28]` |
| 7 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[19,21,27,29]` |
| 8 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[22,23,30,31]` |

Table 5: Wyckoff site: `8e`, site symmetry: `..2'/m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{3}{4}\right]$ | `[1,16,24,25]` |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | `[2,12,20,26]` |
| 3 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | `[3,13,21,27]` |
| 4 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | `[4,10,18,28]` |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 5 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [5,11,19,29] |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$ | [6,15,23,30] |
| 7 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$ | [7,14,22,31] |
| 8 | $\left[0, 0, \frac{1}{4}\right]$ | [8,9,17,32] |

Table 6: Wyckoff site: 8f, site symmetry: 2m'm'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | [1,6,28,29] |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$ | [2,3,31,32] |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$ | [4,5,25,30] |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | [7,8,26,27] |
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | [9,14,20,21] |
| 6 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$ | [10,11,23,24] |
| 7 | $\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$ | [12,13,17,22] |
| 8 | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | [15,16,18,19] |

Table 7: Wyckoff site: 16g, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, -x, \frac{1}{2}\right]$ | [1,8] |
| 2 | $\left[x+\frac{1}{2}, x, 0\right]$ | [2,4] |
| 3 | $\left[-x, \frac{1}{2}-x, 0\right]$ | [3,5] |
| 4 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$ | [6,7] |
| 5 | $\left[-x, x, \frac{1}{2}\right]$ | [9,16] |
| 6 | $\left[\frac{1}{2}-x, -x, 0\right]$ | [10,12] |
| 7 | $\left[x, x+\frac{1}{2}, 0\right]$ | [11,13] |
| 8 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | [14,15] |
| 9 | $[x, -x, 0]$ | [17,24] |
| 10 | $\left[x+\frac{1}{2}, x, \frac{1}{2}\right]$ | [18,20] |
| 11 | $\left[-x, \frac{1}{2}-x, \frac{1}{2}\right]$ | [19,21] |
| 12 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, 0\right]$ | [22,23] |
| 13 | $[-x, x, 0]$ | [25,32] |
| 14 | $\left[\frac{1}{2}-x, -x, \frac{1}{2}\right]$ | [26,28] |
| 15 | $\left[x, x+\frac{1}{2}, \frac{1}{2}\right]$ | [27,29] |
| 16 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, 0\right]$ | [30,31] |

Table 8: Wyckoff site: 16h, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, -x, \frac{3}{4}\right]$ | [1,24] |
| 2 | $\left[x+\frac{1}{2}, x, \frac{1}{4}\right]$ | [2,20] |
| 3 | $\left[-x, \frac{1}{2}-x, \frac{1}{4}\right]$ | [3,21] |

*continued ...*

Table 8

| No. | position | mapping |
|---|---|---|
| 4 | $\left[x+\frac{1}{2},\, x,\, \frac{3}{4}\right]$ | [4,18] |
| 5 | $\left[-x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$ | [5,19] |
| 6 | $\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$ | [6,23] |
| 7 | $\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$ | [7,22] |
| 8 | $\left[x,\, -x,\, \frac{1}{4}\right]$ | [8,17] |
| 9 | $\left[-x,\, x,\, \frac{1}{4}\right]$ | [9,32] |
| 10 | $\left[\frac{1}{2}-x,\, -x,\, \frac{3}{4}\right]$ | [10,28] |
| 11 | $\left[x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$ | [11,29] |
| 12 | $\left[\frac{1}{2}-x,\, -x,\, \frac{1}{4}\right]$ | [12,26] |
| 13 | $\left[x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$ | [13,27] |
| 14 | $\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$ | [14,31] |
| 15 | $\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$ | [15,30] |
| 16 | $\left[-x,\, x,\, \frac{3}{4}\right]$ | [16,25] |

Table 9: Wyckoff site: 16i, site symmetry: .m'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\, y,\, z\right]$ | [1,28] |
| 2 | $\left[\frac{1}{2}-y,\, \frac{1}{4},\, z+\frac{1}{2}\right]$ | [2,31] |
| 3 | $\left[y,\, \frac{1}{4},\, z+\frac{1}{2}\right]$ | [3,32] |
| 4 | $\left[\frac{3}{4},\, -y,\, \frac{1}{2}-z\right]$ | [4,25] |
| 5 | $\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [5,30] |
| 6 | $\left[\frac{1}{4},\, \frac{1}{2}-y,\, z\right]$ | [6,29] |
| 7 | $\left[y+\frac{1}{2},\, \frac{3}{4},\, -z\right]$ | [7,26] |
| 8 | $\left[-y,\, \frac{3}{4},\, -z\right]$ | [8,27] |
| 9 | $\left[\frac{3}{4},\, -y,\, -z\right]$ | [9,20] |
| 10 | $\left[y+\frac{1}{2},\, \frac{3}{4},\, \frac{1}{2}-z\right]$ | [10,23] |
| 11 | $\left[-y,\, \frac{3}{4},\, \frac{1}{2}-z\right]$ | [11,24] |
| 12 | $\left[\frac{1}{4},\, y,\, z+\frac{1}{2}\right]$ | [12,17] |
| 13 | $\left[\frac{1}{4},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | [13,22] |
| 14 | $\left[\frac{3}{4},\, y+\frac{1}{2},\, -z\right]$ | [14,21] |
| 15 | $\left[\frac{1}{2}-y,\, \frac{1}{4},\, z\right]$ | [15,18] |
| 16 | $\left[y,\, \frac{1}{4},\, z\right]$ | [16,19] |

Table 10: Wyckoff site: 16j, site symmetry: ..m

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, x,\, z]$ | [1,16] |
| 2 | $\left[\frac{1}{2}-x,\, x,\, z+\frac{1}{2}\right]$ | [2,12] |
| 3 | $\left[x,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [3,13] |
| 4 | $\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$ | [4,10] |
| 5 | $\left[-x,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [5,11] |
| 6 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, z\right]$ | [6,15] |
| 7 | $\left[x+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$ | [7,14] |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 8 | $[-x,\ -x,\ -z]$ | [8,9] |
| 9 | $\left[x,\ x,\ z+\frac{1}{2}\right]$ | [17,32] |
| 10 | $\left[\frac{1}{2}-x,\ x,\ z\right]$ | [18,28] |
| 11 | $\left[x,\ \frac{1}{2}-x,\ z\right]$ | [19,29] |
| 12 | $\left[x+\frac{1}{2},\ -x,\ -z\right]$ | [20,26] |
| 13 | $\left[-x,\ x+\frac{1}{2},\ -z\right]$ | [21,27] |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [22,31] |
| 15 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [23,30] |
| 16 | $\left[-x,\ -x,\ \frac{1}{2}-z\right]$ | [24,25] |

Table 11: Wyckoff site: 32k, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$ | [2] |
| 3 | $\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [3] |
| 4 | $\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$ | [4] |
| 5 | $\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [5] |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$ | [6] |
| 7 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$ | [7] |
| 8 | $[-y,\ -x,\ -z]$ | [8] |
| 9 | $[-x,\ -y,\ -z]$ | [9] |
| 10 | $\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$ | [10] |
| 11 | $\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [13] |
| 14 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$ | [14] |
| 15 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$ | [15] |
| 16 | $[y,\ x,\ z]$ | [16] |
| 17 | $\left[x,\ y,\ z+\frac{1}{2}\right]$ | [17] |
| 18 | $\left[\frac{1}{2}-y,\ x,\ z\right]$ | [18] |
| 19 | $\left[y,\ \frac{1}{2}-x,\ z\right]$ | [19] |
| 20 | $\left[x+\frac{1}{2},\ -y,\ -z\right]$ | [20] |
| 21 | $\left[-x,\ y+\frac{1}{2},\ -z\right]$ | [21] |
| 22 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [22] |
| 23 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [23] |
| 24 | $\left[-y,\ -x,\ \frac{1}{2}-z\right]$ | [24] |
| 25 | $\left[-x,\ -y,\ \frac{1}{2}-z\right]$ | [25] |
| 26 | $\left[y+\frac{1}{2},\ -x,\ -z\right]$ | [26] |
| 27 | $\left[-y,\ x+\frac{1}{2},\ -z\right]$ | [27] |
| 28 | $\left[\frac{1}{2}-x,\ y,\ z\right]$ | [28] |
| 29 | $\left[x,\ \frac{1}{2}-y,\ z\right]$ | [29] |
| 30 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [30] |
| 31 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [31] |
| 32 | $\left[y,\ x,\ z+\frac{1}{2}\right]$ | [32] |